

Instruments, Not Oracles: Incompleteness, Quantum Logic, and the Limits of Deductive Natural Theology

Abstract

This paper argues that the four principal deductive arguments for God's existence — the Kalam cosmological, contingency, moral, and ontological arguments — share a common dependence on treating logic, modal semantics, and causal principles as transparent windows onto metaphysical structure. Developing a pragmatist, instrumentalist framework, it contends that these formal tools are revisable, domain-relative human instruments shaped by evolutionary and empirical constraints, whose internal structure cannot license metaphysical conclusions about necessary existence. Three formal results independently reinforce the case: Gödel's incompleteness theorems, the non-distributive lattice structure of quantum-mechanical propositions, and the model-theoretic gap between Kripke-frame validity and truth-in-reality.

Keywords: philosophy of religion, natural theology, pragmatism, instrumentalism, Gödel's incompleteness theorems, quantum logic, modal logic, Kripke semantics, ontological argument, Kalam cosmological argument

1. Introduction

Modern analytic natural theology attempts to revive a classical project: demonstrating the existence of God through deductive argument. Craig (2008), Pruss (2009), Koons (1997), and Plantinga (1974) present arguments that purport to be not merely abductive or probabilistic but rationally compelling — arguments whose conclusions follow necessarily from premises defended as self-evident or metaphysically secure. These arguments span different domains — causation, explanation, morality, modality — but they converge on a single structural strategy: they take the internal features of our formal systems (classical logic, modal semantics, the Principle of Sufficient Reason) and treat them as disclosing the deep structure of mind-independent reality.

This paper articulates a systematic challenge to that strategy. Drawing on the pragmatist tradition from Dewey (1938) through Quine (1951) to Putnam (1981, 2004), it develops an instrumentalist framework on which logic, mathematics, and modal semantics are *instruments*: cognitive tools shaped by evolutionary, social, and empirical constraints, whose success is measured by predictive power, coherence, and practical utility rather than by their alleged capacity to mirror a mind-independent metaphysical order. Once these tools are recognized as revisable, domain-relative, and empirically constrained, the

inferential bridge from ‘our logical system licenses X’ to ‘reality necessarily contains X’ collapses — and with it, the ambitions of deductive natural theology.

The paper’s contribution is not merely programmatic. Three bodies of formal results are shown to reinforce the instrumentalist diagnosis in precise and independent ways:

1. **Gödel’s incompleteness theorems** establish that no sufficiently powerful consistent formal system can prove all truths about its intended domain or establish its own consistency from within. This forecloses the possibility that any formal system — including the modal logics on which theistic arguments depend — can serve as a self-certifying mirror of reality.
2. **The non-distributive lattice structure of quantum mechanics**, established by Birkhoff and von Neumann (1936) and reinforced by Bell’s theorem (1964) and the Kochen-Specker theorem (1967), demonstrates that the propositional logic of our best-confirmed physical theory is not classical. This provides concrete, empirically grounded evidence that classical logic is not the universal logic of reality.
3. **The model-theoretic gap** between validity-in-a-Kripke-frame and truth-in-reality — illuminated by Kripke’s completeness theorems (1963) and Putnam’s model-theoretic argument (1980) — shows that theorems of modal logic, including those deployed in ontological arguments, establish at most structural features of formal models, not the existence of entities in the actual world.

Each of these results, taken individually, puts pressure on the realist assumptions undergirding deductive natural theology. Taken together, they constitute a cumulative case that the formal apparatus on which such arguments depend cannot bear the metaphysical weight placed upon it.

Two clarifications are in order. First, the instrumentalist position should not be confused with logical positivism. The positivists held that metaphysical claims are *meaningless* (Ayer 1936, 31–36). The present view is more modest: metaphysical claims may be perfectly meaningful, but formal logic and modal reasoning — being tools with limited and empirically constrained domains of reliability — cannot by themselves establish them. Second, the position is not self-refuting in any obvious sense. The instrumentalist does not claim that ‘instrumentalism is metaphysically necessarily true’; the claim is that our formal tools, assessed by their track record and by the theoretical pressures of incompleteness and quantum mechanics, do not warrant the metaphysical extrapolations that theistic arguments require. This is a claim about the *scope* of certain cognitive instruments, not about the unreliability of all reasoning.

Oppy (2006) and Mackie (1982) have challenged theistic arguments on various internal grounds; Sobel (2004) has subjected several to rigorous logical scrutiny. The present paper identifies a unified structural vulnerability that runs through all four arguments — one that is not a defect in any particular premise but a consequence of the epistemic status of the formal apparatus on which the entire enterprise depends.

The paper proceeds as follows. Section 2 identifies the realist presuppositions shared by deductive natural theology. Section 3 sketches the instrumentalist alternative. Sections 4–6 develop the three formal pressure points: incompleteness, quantum logic, and model theory. Section 7 applies the framework to the four arguments as case studies. Section 8 distills the common failure mode. Section 9 addresses objections. Section 10 concludes.

2. The Realist Presuppositions of Deductive Natural Theology

Despite their diversity, the Kalam cosmological argument, the contingency argument, the moral argument, and the modal ontological argument share four background commitments — commitments that are typically left implicit rather than defended:

1. **Logical realism.** Human logical systems (usually classical logic) track necessary features of reality. The laws of logic are not conventions or cognitive adaptations but genuine constraints on what can and cannot be the case.
2. **Robust modality.** Talk of possibility and necessity corresponds to genuine modal facts. When we say ‘it is possible that God exists,’ we are not merely reporting that a certain sentence is consistent with our axioms; we are making a claim about a mind-independent modal landscape — about what *could* be the case independently of our conceptual frameworks.
3. **Metaphysical causation and explanation.** Causal principles and the Principle of Sufficient Reason (PSR) have metaphysical authority, not merely heuristic value. The claim that ‘whatever begins to exist has a cause’ is not an inductive generalization from experience but a truth about the necessary structure of being.
4. **Abstract structure as ontological guide.** Model-theoretic constructions — possible worlds, accessibility relations, higher-order properties — are treated as revealing the structure of reality itself. The internal features of our formal models are read as evidence about what actually exists.

Each of these commitments is required for the theistic arguments to function. Remove logical realism, and the deductive structure of the arguments loses its metaphysical force. Remove robust modality, and the ontological argument’s key premise (‘it is *possible* that a maximally great being exists’) becomes a claim about our models rather than about reality. Remove the metaphysical status of causation, and the Kalam’s first premise becomes a useful heuristic rather than a necessary truth. Remove the assumption that formal structure tracks ontological structure, and the inferential machinery that transforms abstract logical results into existence claims is disabled.

The instrumentalist challenge targets these commitments at their root.

3. The Instrumentalist Framework

3.1 Logic as tool: from Dewey to Putnam

The instrumentalist tradition offers a fundamentally different picture of logic and formal reasoning. For Dewey (1938, 101–119), logical forms are not descriptions of reality but *operational* — patterns of inquiry that organize experience. Quine's (1951) attack on the analytic-synthetic distinction undermined the idea that logical truths occupy a privileged, unrevisable epistemic category: even the laws of logic are, in principle, subject to revision in light of recalcitrant experience. Putnam (1981, 49–74) developed this into 'internal realism,' on which truth is not correspondence to a scheme-independent world but idealized rational acceptability within a conceptual framework.

On this view, logic, mathematics, and modal semantics are *instruments* shaped by evolutionary, cognitive, and empirical constraints. Their success is measured in terms of predictive power, coherence, and practical utility, not in terms of their alleged ability to disclose the 'deep structure' of reality. Under this view, it is a category mistake to infer ontological necessity from the internal structure of our representational tools.

A natural objection presents itself immediately: if mathematics is merely a tool, why does it work so spectacularly well? Wigner (1960) famously spoke of the 'unreasonable effectiveness of mathematics in the natural sciences,' and apologists have seized on this as evidence that mathematical structure reflects a divine Logos (Craig 2008, 107–108). The instrumentalist reply is that the effectiveness is entirely *reasonable* once one recognizes that our formal tools are products of an ongoing adaptive process. Douglas Adams captured the essential point with characteristic precision: 'Imagine a puddle waking up one morning and thinking, "This is an interesting world I find myself in — an interesting hole I find myself in — fits me rather neatly, doesn't it? In fact it fits me staggeringly well, must have been made to have me in it!"' (Adams 2002, 131). Mathematics is the puddle; the physical universe is the hole. We did not discover a pre-existing mathematical order waiting to be revealed. We *evolved* cognitive tools calibrated to the regularities of this specific physical environment, and we retain and refine exactly those tools that prove effective — discarding the rest. When Euclidean geometry failed to describe curved spacetime, we did not conclude that the universe was defective; we invented non-Euclidean geometry to fit the new shape of the hole. When classical logic failed to describe quantum superposition, we developed quantum logic. The 'fit' between mathematics and physics is not a miracle requiring a divine explanation; it is a survivorship effect, no more mysterious than the fit between a river and its banks.

3.2 The plurality of logics

From an instrumentalist standpoint, the existence of multiple, mutually incompatible logics is expected and unproblematic. Non-classical logics — intuitionistic, relevant, paraconsistent, fuzzy — are internally coherent and useful for different domains (Priest 2008). No single logic has established a claim to be *the* unique logic of reality. Classical

logic works extraordinarily well in many contexts, but its status as metaphysically privileged depends on the very realist assumptions the instrumentalist rejects. As Hacking (1979) argued, asking ‘what is the *correct* logic?’ may be as misguided as asking ‘what is the *correct* geometry?’ — the answer depends on the domain and purpose.

4. Formal Limits I: Gödel’s Incompleteness Theorems

4.1 The theorems stated

Gödel’s two incompleteness theorems (1931) are among the deepest results in mathematical logic. They apply to any formal system S satisfying three conditions: (i) S is *recursively axiomatizable* (there is an effective procedure for listing its axioms); (ii) S is *consistent* (it does not prove both a sentence and its negation); and (iii) S is *sufficiently powerful* (it can represent basic arithmetic — specifically, it extends Robinson arithmetic Q).

First Incompleteness Theorem. If S satisfies (i)–(iii), then there exists a sentence G in the language of S such that: (a) if S is consistent, then S does not prove G ; and (b) if S is ω -consistent (a slightly stronger condition, weakened to mere consistency by Rosser’s strengthening), then S does not prove $\neg G$. G is the *Gödel sentence* for S , and it is constructed via the diagonal lemma to ‘say’ of itself, in effect, ‘I am not provable in S .’ Since G is true (it is indeed unprovable in S , assuming consistency) but not provable, S is *incomplete*: there are truths in its intended domain that it cannot prove.

Second Incompleteness Theorem. If S satisfies (i)–(iii) and is consistent, then S cannot prove its own consistency — that is, S cannot prove the sentence $\text{Con}(S)$, formalized as ‘ $0 = 1$ is not derivable in S .’ The system cannot, from within, certify its own reliability.

4.2 Philosophical implications for natural theology

These results bear on natural theology in two interconnected ways.

First, **no formal system can serve as a self-authenticating mirror of reality.** Apologists who deploy modal logic to establish conclusions about what *must* exist are working within formal systems. Gödel’s results show that such systems are inherently limited: they cannot capture all truths about their intended domains, and they cannot certify their own consistency from within. The move from ‘provable in S5 modal logic’ to ‘metaphysically necessary’ requires treating S5 as complete and reliable with respect to metaphysical modality — precisely the kind of self-grounding that Gödel shows no formal system can provide. As Franzén (2005, 101–120) emphasizes, Gödel’s theorems do not render formal systems useless, but they do establish that their internal deliverances cannot be taken as guarantees about external reality without supplementary, non-formal justification.

Second, **mathematical practice itself is tool-like.** Mathematicians routinely move beyond any given axiom system by adopting new axioms (large cardinal axioms, the axiom of determinacy, Martin’s axiom) to settle previously undecidable questions (Maddy 1997, 73–95). This reflects a dynamic, pragmatic adaptation — the tool is extended when it

proves insufficient — rather than progressive discovery of a fixed realm of Platonic truths. The picture of mathematics as revealing necessary truths that logic can then leverage into existence claims is undermined by the practice of mathematics itself.

4.3 What incompleteness does and does not show

A cautionary note is warranted. Gödel's theorems are frequently misapplied in philosophical contexts (Franzén 2005, 167–172). The present argument does not claim that incompleteness *refutes* the conclusions of theistic arguments. The claim is more targeted but precise.

Consider the modal ontological argument. Its conclusion — 'necessarily, a maximally great being exists' — is derived within a formal system (S5 modal logic augmented with axioms about maximal greatness). The argument's force depends on the assumption that this derivation reveals something about reality: that the proven theorem corresponds to a metaphysical fact. But the second incompleteness theorem tells us that S5 (augmented with any axioms sufficient for basic arithmetic) cannot prove its own consistency. If the system is inconsistent, every formula is provable, including 'necessarily, God exists' — but vacuously. The proponent of the argument must therefore rely on a *meta-theoretic* assurance that the system is consistent, an assurance that, by Gödel's own result, cannot come from within the system itself. This does not make the argument worthless, but it shows that its force is conditional on a consistency assumption that the formal apparatus cannot self-certify. The proponent is not, as is sometimes claimed, extracting a truth about reality from pure logic; the proponent is making a bet on the reliability of a particular formal system, and incompleteness shows that no formal system can validate its own bet.

More generally, the first incompleteness theorem shows that the truths *about* any sufficiently powerful formal system outstrip what the system can prove. There are arithmetical truths — let alone metaphysical ones — that lie beyond the system's reach. The picture of a formal system as a complete catalogue of necessary truths, from which the existence of God can be read off, is precisely what Gödel demolished.

5. Formal Limits II: Quantum Mechanics and the Structure of Propositions

5.1 The Birkhoff-von Neumann lattice

In classical mechanics, the propositions about a physical system (e.g., 'the particle's position is in region R') correspond to measurable subsets of phase space. These form a Boolean algebra — a complemented, distributive lattice — under set-theoretic intersection (conjunction), union (disjunction), and complement (negation). Classical propositional logic, with its distributive laws, is the natural logic of this algebraic structure.

Birkhoff and von Neumann (1936) showed that in quantum mechanics, the corresponding algebraic structure is fundamentally different. The propositions about a quantum system correspond to *closed linear subspaces* of a Hilbert space H (equivalently, to orthogonal

projection operators). These form an *orthomodular lattice* $L(H)$ under the operations of subspace intersection (meet, $\wedge$), closed linear span (join, $\vee$), and orthogonal complement (negation, $\perp$). Crucially, $L(H)$ satisfies the orthomodular law — a weakening of the modular law — but **fails to satisfy distributivity**:

$$a \wedge (b \vee c) \neq (a \wedge b) \vee (a \wedge c), \text{ in general.}$$

5.2 A concrete example

The failure of distributivity is not a mathematical curiosity; it is directly connected to the physical content of quantum mechanics. Consider a spin- $\frac{1}{2}$ particle (e.g., an electron) in a two-dimensional Hilbert space $\mathbb{C}^2$. Let:

- A = ‘spin along the z-axis is $+\hbar/2$ ’ — the one-dimensional subspace $\text{span}\{|\uparrow_z\rangle\}$
- B = ‘spin along the x-axis is $+\hbar/2$ ’ — $\text{span}\{|\uparrow_x\rangle\}$, where $|\uparrow_x\rangle = (1/\sqrt{2})(|\uparrow_z\rangle + |\downarrow_z\rangle)$
- C = ‘spin along the x-axis is $-\hbar/2$ ’ — $\text{span}\{|\downarrow_x\rangle\}$, where $|\downarrow_x\rangle = (1/\sqrt{2})(|\uparrow_z\rangle - |\downarrow_z\rangle)$

Since $|\uparrow_x\rangle$ and $|\downarrow_x\rangle$ form a complete orthonormal basis for $\mathbb{C}^2$, their closed linear span is the entire space: $B \vee C = \mathbb{C}^2 = T$ (the tautologically true proposition). Therefore: $A \wedge (B \vee C) = A \wedge T = A = \text{span}\{|\uparrow_z\rangle\}$.

But $A \wedge B = \{0\}$ and $A \wedge C = \{0\}$, because no non-zero vector is simultaneously an eigenstate of S_z with eigenvalue $+\hbar/2$ and an eigenstate of S_x — the operators do not commute. Hence: $(A \wedge B) \vee (A \wedge C) = \{0\} \vee \{0\} = \{0\} = \perp$ (the trivially false proposition).

The left side is A (a non-trivial proposition); the right side is $\perp$ (a contradiction). Distributivity fails.

5.3 Gleason’s theorem and the non-classicality of quantum probability

Gleason’s theorem (1957) deepens the picture. In a Hilbert space of dimension ≥ 3 , every countably additive probability measure on $L(H)$ — every way of consistently assigning probabilities to quantum propositions — must be of the form $\mu(A) = \text{Tr}(\rho P_A)$ for some density operator ρ . This result has a powerful corollary: there are no two-valued (0 or 1) probability measures on $L(H)$ for dimension ≥ 3 , since no density operator assigns probability 0 or 1 to every projection. In other words, the lattice of quantum propositions does not admit a classical, deterministic probability assignment. The Born rule — the fundamental probability rule of quantum mechanics — is the *unique* probability rule compatible with the Hilbert space lattice structure.

For the philosophy of logic, the significance is this: classical propositional logic presupposes bivalence — every proposition is either true or false. The lattice $L(H)$ does not support bivalent truth-value assignments (for dimension ≥ 3). Whatever one makes of this philosophically, it demonstrates that the classical logical assumption of definite, pre-existing truth values is not a universal feature of reality as our best physics describes it.

5.4 Bell's theorem and contextuality

The non-classicality of the quantum proposition lattice is reinforced by two further results. Bell's theorem (1964) establishes that no local hidden-variable theory can reproduce the statistical predictions of quantum mechanics. The CHSH inequality (Clauser *et al.* 1969) — $|E(a,b) - E(a,b') + E(a',b) + E(a',b')| \leq 2$ — is violated by quantum mechanics, which predicts a maximum value of $2\sqrt{2} \approx 2.83$. This violation has been confirmed to high precision by the loophole-free experiments of Hensen *et al.* (2015), work that contributed to the 2022 Nobel Prize in Physics. Bell's result means that at least one of the following classical assumptions must be abandoned: locality (no faster-than-light causal influence), value-definiteness (all observables have pre-existing definite values), or measurement independence.

The Kochen-Specker theorem (1967) complements Bell by establishing that in a Hilbert space of dimension ≥ 3 , there is no assignment of definite values (0 or 1) to all projection operators that respects the functional relationships between commuting observables and is independent of measurement context. Any hidden-variable theory reproducing quantum predictions must be *contextual*: the value assigned to an observable must depend on which other compatible observables are measured simultaneously.

Together, these results show that the classical assumptions of bivalence, value-definiteness, locality, and context-independence — assumptions silently presupposed by classical logic and by the theistic arguments that deploy it — do not hold universally.

5.5 Scope and caveats

A careful statement of the instrumentalist claim is essential here. The argument is **not** that quantum mechanics refutes classical logic, nor that quantum logic should replace classical logic as the 'true' logic. Putnam (1968) once advanced precisely this claim — that the failure of distributivity in quantum mechanics constitutes an empirical revision of logic, analogous to the revision of Euclidean geometry by general relativity. Putnam subsequently retracted this position, acknowledging in 'The Tale of Quantum Logic' (2005) that quantum logic does not solve the measurement problem and that the case for treating it as a revision of logic *per se* is weaker than he had supposed.

The present paper's claim is more modest and, for that reason, more defensible. The claim is not that classical logic is false but that it is **domain-limited**: it is one formal tool among several, extraordinarily successful in many contexts but not universally applicable. Our best-confirmed physical theory employs a propositional structure that is non-distributive, non-Boolean, and contextual. This empirical fact — regardless of its ultimate philosophical interpretation — undermines the assumption that classical logic is the unique and unrevisable logic of reality, an assumption on which deductive natural theology critically depends.

6. Formal Limits III: Model Theory and the Gap Between Validity and Truth

6.1 Kripke semantics and modal completeness

Kripke (1963) proved completeness theorems for a family of propositional modal logics relative to classes of relational frames $\langle W, R \rangle$, where W is a set of ‘possible worlds’ and R is a binary accessibility relation:

System	Constraint on R	Key axiom
K	none	—
T	reflexive	$\Box\phi \rightarrow \phi$
S4	reflexive, transitive	$\Box\phi \rightarrow \Box\Box\phi$
S5	equivalence relation	$\Diamond\phi \rightarrow \Box\Diamond\phi$

A formula ϕ is *provable in L* if and only if ϕ is *valid in every frame in the corresponding class*. This is a powerful and beautiful result. But it establishes a correspondence between provability and **frame-validity**, not between provability and **metaphysical truth**. A theorem of S5 — including, say, ‘if possibly necessarily p , then necessarily p ’ — is guaranteed to hold in every model built on an equivalence-relation frame. Whether it holds *in reality* depends on whether reality’s modal structure is correctly represented by an equivalence-relation frame, and that is a substantive metaphysical question that the completeness theorem does not address.

6.2 Gödel’s ontological proof and modal collapse

This point is sharpened by considering Gödel’s own ontological proof (Gödel 1970/1995), a higher-order modal argument that derives the necessary existence of a God-like being from axioms about ‘positive properties.’ Sobel (1987) demonstrated that Gödel’s axioms entail **modal collapse**: every true proposition is necessarily true ($\Box\phi \leftrightarrow \phi$ for all ϕ) — a result computationally verified by Benz Müller and Woltzenlogel Paleo (2014) using the Isabelle/HOL theorem prover. Anderson (1990) and Fitting (2002) offered emendations to avoid the collapse, trading on subtle distinctions between extensional and intensional readings of the axioms.

The significance for the present argument is this: within the formal system, the proof is valid. But its validity generates absurd consequences (the elimination of all contingency) unless the axioms are carefully adjusted — and even then, the proof establishes only that a model exists in which its conclusion holds, not that its conclusion describes reality. The gap between intra-systematic validity and extra-systematic truth is not a minor technicality; it is the central issue.

6.3 Non-isomorphic models and the underdetermination of reference

The gap between validity and truth is further widened by a basic fact of model theory: theories typically have multiple, non-isomorphic models. A theorem that holds ‘in all models’ holds in each of these distinct structures. But which (if any) corresponds to reality? Model-theoretic truth is truth-in-all-models, not truth-about-reality. When the ontological argument establishes that ‘in every S5-model satisfying axiom A, a necessary being exists,’ it has established a structural fact about a class of mathematical objects. The inference from this structural fact to the existence of an entity in the actual world requires the additional, non-formal assumption that reality is (or is correctly represented by) one of these models — an assumption the formal apparatus cannot justify.

6.4 Putnam’s model-theoretic argument

Putnam (1980) pressed this point to its sharpest form. Using the downward Löwenheim-Skolem theorem — which guarantees that any first-order theory with an infinite model has a countable model — he argued that even an ‘ideal’ theory satisfying all operational and theoretical constraints cannot fix a unique intended interpretation. A theory that quantifies over uncountably many real numbers has, by Löwenheim-Skolem, a countable model in which all its sentences are true. The objects in this model are not ‘really’ real numbers, yet the theory cannot tell the difference from within. Adding causal-reference constraints to the theory is, Putnam argued, ‘just more theory’ — more sentences subject to the same model-theoretic permutation.

If this is correct for scientific theories, it applies *a fortiori* to modal-logical theories deployed to establish existence claims about God: the formal apparatus cannot, by itself, guarantee that its theorems latch onto reality rather than an artefact of the model. The gap between what a formal system proves and what reality contains is not a gap that more formalism can bridge.

7. Case Studies: The Four Arguments Under Instrumentalism

With the formal groundwork in place, the four principal arguments can be evaluated as case studies illustrating a single pattern of overreach.

7.1 The Kalam cosmological argument

Craig’s Kalam (Craig 2008, 111; Craig and Sinclair 2009) runs: (1) Whatever begins to exist has a cause; (2) The universe began to exist; (3) Therefore, the universe has a cause — and the cause is argued to be timeless, spaceless, immaterial, and personal. Premise (1) is defended as a metaphysical necessity. On the instrumentalist view, it is at most a high-confidence inductive generalization in the macroscopic domain — a domain where classical logic and classical causation are well-calibrated tools. Quantum phenomena (radioactive decay, vacuum fluctuations) show that our best physical theories do not uniformly support the universality of classical causal principles (Maudlin 2011, 1–22).

Extending causal intuitions honed inside the universe to the putative origin of space-time is to project a local tool far beyond its domain of calibration.

Craig's inference from 'a cause' to 'a personal, timeless creator' compounds the problem. As Morriston (2000) argues, genuine personal agency requires temporal deliberation — but a timeless being cannot deliberate in any recognizable sense. The inference from timelessness to personhood relies on modal intuitions about what a timeless cause 'must' be like — intuitions grounded in our experience of embodied agents in time, with no independent warrant for extension to extra-temporal entities.

7.2 The contingency argument

The contingency argument (Pruss 2009, 24–25) relies on the PSR — that every contingent fact has an explanation — to argue for a necessary being. On the instrumentalist view, the PSR is a methodological maxim (*wherever possible, seek explanations*), not a law of being. Its impressive success within science does not license the metaphysical thesis that no brute facts exist (van Inwagen 1983, 202–204; Oppy 2006, 95–100). Once the PSR is recognized as a heuristic, the argument establishes at most that a necessary being is one *possible* terminus of explanation, not that it is a logically compelled one. The phrase 'necessary being' itself, on the instrumentalist view, names a position within a conceptual scheme — a statement's necessity is relative to a chosen logic, set of axioms, or framework — and the move from intra-systematic necessity to metaphysical necessity is precisely the use-mention error the formal results of Sections 4–6 diagnose.

7.3 The moral argument

Craig's moral argument (Craig 2008, 172–176) contends that if God does not exist, objective moral values do not exist; but objective moral values do exist; therefore God exists. The argument trades on an equivocation between two senses of 'objective': stance-independent (moral truths hold regardless of all attitudes — requiring a transcendent ground) and robustly inter-subjective (moral norms are stable under wide scrutiny and embedded in successful social practices). Premise (2) draws its intuitive force from everyday moral experience, which is well explained by the second, more modest sense. But premise (1) requires the first, metaphysically loaded sense to do its work.

The instrumentalist framework provides a natural account of moral objectivity in the second sense. Moral norms can be *highly robust and non-arbitrary* — arising from the interaction of evolved social agents under constraints of cooperation, vulnerability, and limited resources (Street 2006; Joyce 2006; Wielenberg 2014) — without requiring a divine legislator. The Euthyphro dilemma (Plato, *Euthyphro* 10a; Quinn 1978) provides independent pressure: if goodness is constituted by divine command, morality is arbitrary; if God commands what is independently good, goodness is independent of God. Craig's identification of God with the Good itself (Craig 2008, 181–183) is a metaphysical stipulation that does not follow from the argument's structure — it is one more instance of reifying a conceptual role into an entity.

7.4 The ontological argument

Plantinga's modal ontological argument (1974, 213–217) reasons that if it is possible that a maximally great being exists, then (via S5) such a being exists in the actual world. The instrumentalist response draws directly on Sections 4–6. The key premise — that maximal greatness is *possible* — is defended as conceptually coherent. But conceptual coherence within a modal system (consistency with the axioms and rules of S5) is not equivalent to genuine metaphysical possibility (Oppy 1995, 72–85). Kripke's completeness theorem guarantees that if the premise is consistent with S5, there is an S5-frame in which it holds. It does not guarantee that reality is such a frame. The choice of S5 itself is a modelling decision: S5 corresponds to frames where the accessibility relation is an equivalence relation, yielding the powerful axiom $\Diamond\Box\phi \rightarrow \Box\phi$. On a weaker system (K, T, S4), the argument fails. The question of which modal logic correctly captures metaphysical modality is a substantive philosophical question that cannot be settled by the arguments that presuppose the answer (Divers 2002, 155–170; Sobel 2004, 80–100).

The history of Gödel's own ontological proof (Section 6.2) is instructive here. Within a carefully constructed higher-order modal system, Gödel derived the necessary existence of a God-like being — an impeccable formal achievement. Yet the proof's axioms generated modal collapse (every truth is a necessary truth), a consequence so absurd that the proof's proponents have spent decades attempting repairs (Anderson 1990; Fitting 2002). The lesson is not that formal proofs are useless but that formal validity within a system is a weaker guarantee than it appears. The internal coherence of a formal derivation does not protect against metaphysical absurdity unless the system's axioms are independently shown to track reality — and that showing cannot come from within the system itself.

Plantinga (1974, 220–221) himself acknowledged that the modal argument is not a proof in the strong sense; it succeeds only if the key premise is granted. The instrumentalist challenge goes further: it questions the framework within which the premise is assessed.

8. The Common Failure Mode

Across all four arguments, a single four-step pattern emerges:

1. **Domestication.** Start with intuitions and inferential habits that function well within certain domains: causal reasoning in everyday life, explanation-seeking in science, moral judgment in social contexts, conceptual analysis in philosophy.
2. **Formalization.** Codify these habits into formal tools: classical logic, S5 modal semantics, the PSR, possible-worlds frameworks, higher-order property theories.
3. **Reification.** Treat these tools as if they were transparent mirrors of metaphysical structure — as if the internal features of the formal system directly reveal what reality must contain.

4. **Extraction.** Use the internal structure of the reified tools to ‘read off’ the existence and attributes of a theistic God.

The instrumentalist framework blocks step (3). The formal results of Sections 4–6 provide three independent reasons for the block: Gödel shows that formal systems cannot certify their own completeness or correspondence to reality; quantum mechanics shows that the propositional logic of our best-confirmed physical theory is not classical; and model theory shows that the gap between validity-in-a-frame and truth-in-reality cannot be bridged by formal means alone.

Once step (3) is blocked, the bridge from ‘our logic says...’ to ‘reality must contain...’ is broken. The apologetic arguments may illuminate how certain theists organize their conceptual schemes, but they cannot constrain what reality must be like.

9. Objections and Replies

9.1 The self-refutation objection

Objection: The instrumentalist uses logic to argue against the metaphysical reliability of logic. Is this not self-defeating?

Reply: The argument is not that logic is *unreliable* but that it is *domain-limited*. The instrumentalist uses classical logic in the domain where it is well-calibrated (ordinary reasoning, philosophical argumentation) while denying that its internal structure can be projected onto domains where its reliability has not been established (the origin of the universe, the nature of necessary being, the structure of all possible worlds). This is no more self-defeating than a cartographer who uses a Mercator projection for navigation while insisting that it distorts the relative sizes of landmasses. The tool is useful; the error lies in treating its artefacts as features of the territory.

9.2 The ‘proves too much’ objection

Objection: If instrumentalism about logic undermines theistic arguments, does it not also undermine science? Scientific theories depend on logic and mathematics. If formal systems cannot tell us about reality, how can physics do so?

Reply: The instrumentalist distinguishes between formal tools that are **empirically constrained** and formal tools that are **projected beyond all empirical constraint**. Scientific theories are successful precisely because they are continually tested against observation, revised in light of recalcitrant data, and evaluated by predictive track record. The formal apparatus of physics earns its credentials through this ongoing empirical dialogue. Theistic arguments, by contrast, deploy formal tools in a domain — the existence of a necessary being, the origin of all contingent reality — where no empirical feedback is available. The instrumentalist objection is not to the use of logic and mathematics *per se* but to the metaphysical extrapolation of formal results into domains where the tools’ reliability has not been independently established. Van Fraassen’s (1980) constructive empiricism makes a structurally parallel point about scientific theories: one can accept a

theory as empirically adequate without committing to the literal truth of its claims about unobservable entities.

9.3 The indispensability argument

Objection: Mathematics is indispensable to our best scientific theories. By the Quine-Putnam indispensability argument, we ought to be ontologically committed to the mathematical entities those theories quantify over. If mathematical entities are real, mathematical truths are genuine truths about reality, and the instrumentalist position collapses.

Reply: The indispensability argument is contested on multiple fronts (Field 1980; Maddy 1997, 133–157). But even granting it for the sake of argument, it does not rescue the theistic project. The mathematical structures to which science is arguably committed are empirically constrained structures — they earn their ontological credentials through their indispensable role in empirically successful theories. Modal-logical structures deployed in ontological arguments have no comparable empirical grounding. The indispensability argument, if sound, supports realism about the mathematics of physics, not realism about the metaphysical deliverances of S5 modal logic.

9.4 The ‘unreasonable effectiveness’ objection

Objection: Wigner (1960) observed that mathematics is unreasonably effective in the natural sciences — its applicability to the physical world far exceeds what any instrumentalist account would predict. This ‘unreasonable effectiveness’ is best explained by the hypothesis that mathematical structure is woven into the fabric of reality, suggesting a rational, mathematical creator. If mathematics tracks reality so precisely, the instrumentalist has no explanation for the fit, and the realist assumption underlying theistic arguments is vindicated.

Reply: The effectiveness of mathematics is entirely reasonable once we recognize that it is the product of an iterative, evolutionary process of tool-refinement — and that we are subject to a powerful survivorship bias in our assessment of the ‘fit.’

Consider: the history of mathematics and physics is littered with discarded tools — Ptolemaic epicycles, phlogiston theory, Euclidean spacetime, classical deterministic trajectories for quantum particles. These tools were once taken to be perfectly adequate descriptions of reality; they were abandoned when the shape of the evidential hole changed. We do not marvel at their failure; we simply forget them. What remains — the mathematics that *currently* works — creates the illusion of an astonishing, pre-established harmony between mind and world. But this is precisely the puddle’s error (Section 3.1): we are looking at a system that has spent millennia conforming itself to the contours of empirical reality, and we are marvelling at the fit as though reality had been designed to match our mathematics.

The apologist's inference from the effectiveness of mathematics to a divine mathematician is a textbook instance of the paper's four-step failure mode. (1) We develop mathematical tools to cope with physical regularities (*domestication*). (2) We formalize these tools into abstract systems (*formalization*). (3) We forget the adaptive, empirical origins and marvel at the 'unreasonable' precision of the fit, treating the tools as if they were windows onto a pre-existing mathematical order (*reification*). (4) We conclude that a divine intelligence must have designed the mathematical structure of reality (*extraction*). The puddle analogy dissolves step (3) by revealing that the 'fit' is not evidence of design but of adaptation: the mathematics fits reality because we have spent centuries reshaping it until it does.

9.5 Plantinga's evolutionary argument against naturalism

Objection: Plantinga (1993, ch. 12) argues that the conjunction of naturalism and evolution is self-defeating: if our cognitive faculties arose by unguided natural selection, we have no reason to trust their reliability, since selection acts on behaviour, not on truth. If this is correct, the instrumentalist's reliance on evolutionary and naturalistic considerations to ground the tool-based view undermines itself.

Reply: Fitelson and Sober (1998) identify several weaknesses in the EAAN. The most pertinent here is the **total evidence** problem: Plantinga assesses $P(R \mid N\&E)$ — the probability that our faculties are reliable given naturalism and evolution — in abstraction from our actual cognitive track record. Once one conditionalizes on the specific features of our cognitive architecture, its environmental success, and the intersubjective coordination it enables, the probability assessment shifts substantially. Natural selection has grip on content accuracy at least in domains directly relevant to survival, and the tight coupling between perception and behaviour in evolved systems provides selective pressure for truth-tracking in those domains. More fundamentally, the EAAN, even if sound, would generate a symmetrical problem for the theist who claims that God designed our cognitive faculties for reliability — since many human cognitive biases are well-documented, the blanket claim of divinely guaranteed reliability is empirically strained. The instrumentalist need only claim that our cognitive tools are *locally* reliable — well-calibrated in their proper domains — which is precisely the thesis being defended.

10. Conclusion

This paper has argued that deductive natural theology rests on a set of realist presuppositions about logic, modality, and causation that are increasingly difficult to sustain in the light of three bodies of formal results: Gödel's incompleteness theorems, the non-classical structure of quantum-mechanical propositions, and the model-theoretic gap between formal validity and metaphysical truth. An instrumentalist framework — on which logic, mathematics, and modal semantics are revisable, domain-relative cognitive tools — provides a principled basis for recognizing the limits of these tools without dismissing their extraordinary utility within their proper domains.

The four principal deductive arguments for God's existence — the Kalam cosmological, contingency, moral, and ontological arguments — all instantiate a single pattern: the

domestication of successful local practices into formal tools, the reification of those tools as metaphysical mirrors, and the extraction of existence claims from the tools' internal structure. The instrumentalist framework blocks the reification step, and the formal results surveyed here provide three independent reasons for the block.

The implication is not that these arguments are worthless — they remain illuminating as explorations of the structure and limits of human reasoning. Nor is it that theism is false. The implication is that the ambition of *demonstrating* God's existence through deductive-metaphysical means rests on an epistemology of logic and modality that is, by contemporary standards, untenable. The arguments are not refuted in the sense that their conclusions are shown to be false; they are shown to be *non-compulsory*, conditional on metaphysical assumptions about the reality-tracking power of formal systems that the instrumentalist has principled — and now formally reinforced — reasons to reject. Theism, if it is to be rationally maintained, must seek grounding in other kinds of consideration — empirical, experiential, existential, or pragmatic — rather than in allegedly irresistible metaphysical proofs.

Notes

1. The instrumentalist tradition is broader than can be surveyed here. In addition to the pragmatist lineage (Dewey, Quine, Putnam), relevant contributions include van Fraassen's (1980) constructive empiricism, Field's (1980) nominalism about mathematics, and Blackburn's (1993) quasi-realism about various domains including modality. The present paper draws on their shared insight that the success of a representational system does not entail that its internal structure corresponds to the structure of reality.
2. The diagonal lemma, which underlies the construction of the Gödel sentence, works as follows. For any formula $\phi(x)$ with one free variable, there exists a sentence G such that $S \vdash G \leftrightarrow \phi(\ulcorner G \urcorner)$, where $\ulcorner G \urcorner$ is the Gödel number of G . Setting $\phi(x) = \neg \text{Prov}_S(x)$ — 'x is not the Gödel number of a theorem of S' — yields the Gödel sentence G that 'says' of itself: 'I am not provable in S.'
3. The orthomodular law, which $L(H)$ satisfies in place of distributivity, states: if $a \leq b$, then $b = a \vee (b \wedge a^\perp)$. This is strictly weaker than modularity ($a \vee (b \wedge c) = (a \vee b) \wedge c$ whenever $a \leq c$), which is in turn strictly weaker than distributivity. The hierarchy is: Boolean (distributive) $\supset$ modular $\supset$ orthomodular.
4. Plantinga (1974, 220–221) acknowledged that the modal ontological argument is not a proof in the strong sense — it establishes its conclusion only given the key premise that maximal greatness is possible. The argument's force is thus conditional on the metaphysical framework within which that premise is assessed, which is precisely the framework the instrumentalist challenges.

5. Gleason's theorem (1957) further constrains the non-classicality of quantum probability: in a Hilbert space of dimension ≥ 3 , every countably additive probability measure on $L(H)$ is of the form $\mu(A) = \text{Tr}(pP_A)$ for some density operator p . This rules out two-valued (classical, deterministic) probability measures, reinforcing the Kochen-Specker result.
6. Gödel's ontological proof circulated privately from 1970 and was published posthumously in his *Collected Works*, vol. 3 (1995). Sobel's modal collapse result appeared in 'Gödel's Ontological Proof,' in *On Being and Saying: Essays for Richard Cartwright* (MIT Press, 1987). Anderson's emendation — replacing the definition of God-likeness so that a God-like being *necessarily* has all positive properties — is in 'Some Emendations of Gödel's Ontological Proof,' *Faith and Philosophy* 7 (1990), 291–303. Fitting's intensional reformulation is in *Types, Tableaus, and Gödel's God* (Kluwer, 2002).

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